

Theorem 7': rational-derived quartics with a repeated
root

Blueprint of the Lean formalization

Hiroki Fukui

July 21, 2026

Chapter 1

Theorem 7'

This blueprint documents the Lean 4 formalization of Theorem 7': the unconditional repair of the classification of rational-derived quartics with a repeated root (Theorem 7 of Buchholz–MacDougall, J. Number Theory 81 (2000)). Every node below is formalized; the master statement `Thm7Statement.thm7prime` has axiom footprint exactly `[propext, Classical.choice, Quot.sound]`.

1.1 The normalized family and the gates

Definition 1.1 (Normalized $p(2, 1, 1)$ family). For $a \in \mathbb{Q}$ let $q_a(x) = x^2(x - 1)(x - a)$, the normalized quartic with root multiset $\{0, 0, 1, a\}$; it has genuine type $p(2, 1, 1)$ iff $a \notin \{0, 1\}$.

Definition 1.2 (Gates). $G_1(a)$: $9a^2 - 14a + 9$ is a rational square. $G_2(a)$: $9a^2 - 6a + 9$ is a rational square. (Formalized as `Gate1` and `Gate2`.) These are the discriminant conditions for q'_a and q''_a to split over $\mathbb{Q}$.

Definition 1.3 (Rational-derivedness, gate form). $\text{RD}(a)$: $a \neq 0$, $a \neq 1$, $G_1(a)$ and $G_2(a)$.

Lemma 1.4 (Splitting under the gates). Under G_1 resp. G_2 , the derivatives q'_a and q''_a split with explicit rational roots. (Formalized as `dq1_splits` and `dq2_splits`.)

Proof. Quadratic formula; a one-line `linear_combination`. □

Lemma 1.5 (Normalization of a general $p(2, 1, 1)$ translate). For $r_1 \neq c$, the quartic $(x - c)^2(x - r_1)(x - r_2)$ pulled back along $x = c + (r_1 - c)u$ equals $(r_1 - c)^4 q_a(u)$ with $a = (r_2 - c)/(r_1 - c)$.

Proof. A ring identity. □

1.2 The curve and the a -map

Definition 1.6 (The curve E). E : $z^2 = w^3 + 12w^2 - 108w = w(w - 6)(w + 18)$, Cremona 576i2. The membership predicate is `OnE`; the mathlib Weierstrass model is `E576i2`, tied to `OnE` by `OnE_iff_equation`, with $\Delta \neq 0$ proved in `E576i2_elliptic`.

Definition 1.7 (The a -map). $a(w, z) = \frac{9(2w + z - 12)(w + 2)}{(z - w - 18)(8w + z)}$, written with the on-curve reduced denominator $D(w, z) = (7w - 18)z + w^3 + 4w^2 - 252w$ (`D_reduce`, `aNum`, `aDen`).

1.3 The statement

Definition 1.8 (Theorem 7', the statement). *Thm7' is the conjunction of the backward direction (**Thm7Backward**: every a -map value on $E(\mathbb{Q})$ away from the denominator locus and from $\{0, 1\}$ satisfies RD) and the forward direction (**Thm7Forward**: every a with $\text{RD}(a)$ arises as such a value).*

1.4 The backward direction

Lemma 1.9 (Cleared gate identities). *On E , cleared of denominators, $9a^2 - 14a + 9$ and $9a^2 - 6a + 9$ (for $a = a(w, z)$) are squares of explicit rational functions; the certificates are single **linear_combination** steps (**gate1_cleared**, **gate2_cleared**), packaged in division form as **gate1_sq**, **gate2_sq**.*

Proof. Sage-lifted cofactor certificates, checked by the kernel. □

Lemma 1.10 (Exceptional locus). *On $E(\mathbb{Q})$, the side factor $w(w + 18)(w^2 - 63w + 486)$ vanishes with $D(w, z) \neq 0$ only at $(54, 432)$ and $(9, -27)$; both map to $a = 77/90$, where the gates hold with witnesses $19/10$, $97/30$ (**gates_at_77_90**).*

Proof. Finite case analysis. □

Theorem 1.11 (Backward direction). *Thm7Backward holds.*

Proof. Generic branch by Lemma 1.9, exceptional branch by Lemma 1.10. No rank input. □

1.5 The forward direction

Definition 1.12 (The inverse map). *Given gate witnesses (r, s) for a , set $w = 18(15a - 6r - s - 13)/(3r - 7s + 12)$ and determine z linearly from $a \cdot D(w, z) = 9(2w + z - 12)(w + 2)$; the denominators involve $K(a, r, s) = 105a^2 - 45ar - 238a + 51r + 16s + 105$ and the sign-selection sextic $P(a, r, s)$ (**wNp**, **wDp**, **zNp**, **zDp**, **K1p**, **adP6**). Discovered by Gröbner elimination.*

Lemma 1.13 (Cleared inverse identities). *Modulo the two gate relations, the inverse point lies on E (**curve_cleared**) and the pulled-back denominator equals $-62208 P(a, r, s)$ up to the chart clearing (**aden_cleared**).*

Proof. Sage-lifted cofactor certificates, checked by the kernel. □

Lemma 1.14 (96 is not a rational square). *For all $x \in \mathbb{Q}$, $x^2 \neq 96$.*

Proof. Via **Rat.sqrt**. □

Lemma 1.15 (Denominator nonvanishing, all signs). *If the gates hold and $a \neq 0$, then $3r - 7s + 12 \neq 0$ and $K(a, r, s) \neq 0$ for every choice of signs of (r, s) : the eliminants are $a^2(225a^2 - 210a + 238)$ with $225a^2 - 210a + 238 = (15a - 7)^2 + 189 > 0$, and $(3a - 5)(5a - 3)a^2(225a^2 - 210a + 238)$, whose linear branches force $(3r)^2 = 96$ resp. $(5r)^2 = 96$ (**wD_ne**, **K1_ne**).*

Proof. Eliminant certificates plus Lemma 1.14 and positivity. □

Lemma 1.16 (Sign cover). *Under the gates, at least one of the four sign choices keeps $P(a, \pm r, \pm s) \neq 0$: the ideal generated by the gate relations and all four conjugates is the unit ideal, and the Nullstellensatz certificate is checked by the kernel.*

Proof. A single `linear.combination` deriving $1 = 0$ from the contrary. $\square$

Lemma 1.17 (Forward core). *Under the gates and the three nonvanishing conditions, the inverse point lies on $E(\mathbb{Q})$, avoids the a -map denominator locus, and maps back to a .*

Proof. Assemble Lemma 1.13 with the chart clearings. $\square$

Theorem 1.18 (Forward direction). *`Thm7Forward` holds. No rank input.*

Proof. Choose signs by Lemma 1.16; apply Lemma 1.17. $\square$

1.6 The main theorem

Theorem 1.19 (Theorem 7', unconditional). *`Thm7'` holds, with axiom footprint exactly `[propext, Classical.choice, Quot.sound]`.*

Proof. `(thm7_backward, thm7_forward)`. $\square$

Remark 1.20 (The disclosed rank axiom). *The development retains one disclosed, instance-localized axiom (`rank_E576i2`: the Mordell–Weil rank of `576i2` is 1, verified unconditionally in two independent systems), consumed only by two auxiliary declarations from an earlier conditional architecture. The main theorem does not depend on it, and the kernel’s per-declaration axiom report documents this separation.*